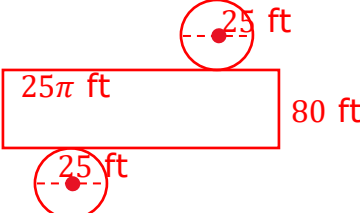


Assume a silo on a farm is in the shape of a right circular cylinder.

1. Complete the table to find the surface area of a silo whose diameter is 25 feet (ft.) with a height of 80 ft. Use 3.14 for π . Round your answer to the nearest hundredth.

Draw a net and label its dimensions (Include units.)			
Determine the area of one circular base. (Include units.)	$A = (3.14)(12.5)^2 \approx 490.63 \text{ ft}^2$		
Find the sum of the areas of both circular bases. (Include units.)	$490.63 + 490.63 = 981.26 \text{ ft}^2$		
Find the lateral area. (Include units.)	$A = (3.14)(25)(80) \approx 6,280 \text{ ft}^2$		
Find the sum of the areas for all the parts. (Include units.)	Sum of the Areas of the Circular Bases	+	Lateral Area
	981.26	+	6280
	$\approx \underline{7,261.26 \text{ ft}^2}$		

2. What is the surface area of the silo?
 $7,261.26 \text{ ft}^2$

James is repainting 3 stools in the shape of a right circular cylinder. He needs to determine how much paint to buy at the hardware store. Each stool has different dimensions in the table. Find the amount of paint needed. Use $\frac{22}{7}$ for π and round to the nearest hundredth.

Stool Dimensions	Work	Amount of Paint Needed
3. Stool 1 $d = 8$ in. $h = 26$ in.	$2(4^2)\left(\frac{22}{7}\right) + \left(\frac{22}{7}\right)(26)(8)$ $100.5714 + 653.7143$	754.29 in ²
4. Stool 2 $d = 14$ in. $h = 36$ in.	$2(7^2)\left(\frac{22}{7}\right) + \left(\frac{22}{7}\right)(14)(36)$ $308 + 1,584$	1,892 in ²
5. Stool 3 $r = 18$ in. $h = 30$ in.	$2(18^2)\left(\frac{22}{7}\right) + \left(\frac{22}{7}\right)(36)(30)$ $2,036.5714 + 3,394.2857$	5,430.86 in ²

Cynthia wants to send a right circular cylinder candle to a friend. When she went to the shipping store, the store explained that the candle must be wrapped in paper. Find the amount of paper needed for the candle. The candle has a diameter of 10 inches (in.) and a height of 8.5 in. Use 3.14 for π .

- How much paper is needed to wrap 1 candle?
 $A = 2(3.14)(5^2) + (3.14)(10)(8.5) \approx 423.9$ in²
- How much paper is needed if Madeline decides to ship 3 more candles with the same dimensions to her friends?
 $423.9 + 3(423.9) = 1,695.6$ in²

Griffin wanted to wrap a drum in the shape of a right circular cylinder for his best friend's birthday. He wanted to make sure that he bought enough wrapping paper. The drum has a diameter of 12 inches (in.) and a height of 20 in.

$$(2 \cdot \pi \cdot 12^2) + 2 \cdot \pi \cdot 12 \cdot 20$$

$$288\pi + 480\pi$$

$$768\pi$$

- Is Griffin correct? If not, circle his error above.
No
- Write a note to Griffin explaining his error.
Used 12 for r instead of 6 & multiplied area of rectangle by 2.
- Correct Griffin's work.
 $2\pi(6^2) + \pi(12)(20)$
 $72\pi + 240\pi$
 312π